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Virulence characterization of wheat stripe rust population in China in 2023

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Abstract

Wheat stripe rust, caused by *Puccinia striiformis* f. sp. *tritici* (Pst), significantly impacts global wheat production, including China. Monitoring Pst virulence is crucial for disease management. This study used 18 *Yr* single-gene isogenic lines of wheat to characterize the virulence of 2023 Pst collections in China. A total of 103 isolates were collected from 11 provinces, and 61 races were identified. Frequencies of virulence to the 18 *Yr* genes varied from 0% to 95.1%, with low frequencies (defined as less than 10%) to *Yr85* (7.8%) and *Yr10* (8.7%); moderate frequencies (10%–40%) to *Yr24* (15.5%), *Yr32* (16.5%), *YrSP* (37.9%) and *Yr76* (37.9%); and high frequencies (>40%) to *Yr7* (47.6%), *Yr1* (68.0%), *Yr9* (69.9%), *Yr27* (69.9%), *Yr17* (79.6%), *Yr6* (89.3%), *YrExp2* (92.2%), *Yr43* (93.2%), *Yr8* (95.1%) and *Yr44* (95.1%). *Yr6*, *Yr8*, *Yr17*, *Yr27*, *Yr43*, *Yr44* and *YrExp2* have almost lost their effectiveness in all provinces, while *Yr10*, *Yr24*, *Yr32* and *Yr85* are effective in Hebei, Shandong, Shaanxi, Gansu and Anhui. No isolate was virulent to *Yr5* or *Yr15*, indicating their effectiveness against Pst populations in China. The Pst isolates had a wide virulence spectrum ranging from 3 to 16 virulence factors, with an average of 9.5. Comparing 2023 to 2021, the PstCN-040 and races sharing virulence to *Yr1*, *Yr6*, *Yr7*, *Yr8*, *Yr9*, *Yr17*, *Yr43*, *Yr44* and *YrExp2* predominated, with frequencies 16.4% and 25.2%. This study provides valuable information on the virulence composition of Pst populations in China, helping to guide the development of genetic resistance for wheat in China.

KEYWORDS
Puccinia striiformis f. sp. *tritici*, race identification, virulence, wheat stripe rust, *Yr* gene

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1 | INTRODUCTION

Stripe (yellow) rust, caused by *Puccinia striiformis* f. sp. *tritici* (Pst), is one of the most destructive diseases of wheat worldwide, and has been reported in more than 60 countries (Chen, 2020). Stripe rust can cause 5%–25% yield loss in normal years, and potentially up to 50% loss or even no yield in severe epidemics (Bouvet et al., 2022; Chen, 2005; Wellings, 2011). China is one of the world's major wheat-producing countries. Since 1950, eight severe epidemics of wheat stripe rust have been documented, resulting in a cumulative loss of 14 billion kg of wheat (Chen et al., 2013, 2014; Kang et al., 2015; Liu et al., 2022).

To control stripe rust, fungicides are often applied, but enormous efforts have been made in the development of resistant wheat cultivars, the use of which is considered the most economical, environmentally friendly and effective approach. Eighty-six officially named yellow rust (*Yr*) resistance genes and many quantitative trait loci (QTLs) have been identified from germplasm of wheat and its relatives worldwide (Zhu et al., 2023). However, Pst has the ability to evolve rapidly to change its virulence, circumventing race-specific resistance and causing epidemics facilitated by its long-distance airborne transmission (Chen et al., 2013; Wang et al., 1988). The number of new races in Pst populations around the world and their ability to defeat more resistance genes have been well documented (Wan, Muleta, et al., 2017). For example, the emergence of *Yr9*-virulent Pst races has rendered more than 80% of Chinese wheat cultivars susceptible (Wan et al., 2004). In Russia, *Yr9* was widely used for over half a century but has lost its effectiveness against stripe rust (Shaydayuk & Gultyaeva, 2020). In the United States, the number of new races detected annually after 2000 was higher than that before 2000 (Wan & Chen, 2014). Similarly, in Canada, races virulent to *Yr8* and *Yr9* appeared in 2000 and a shift to increased virulence was reported in subsequent years (Ghanbarnia et al., 2021). Therefore, monitoring the virulence frequency, geographical distribution and race composition of Pst populations is vital for breeding resistant wheat cultivars and managing the disease.

Race identification is the most direct and accurate method for monitoring the virulence of Pst populations (Wan, 2003). For this, a set of commercialized wheat cultivars or lines from local regions or countries are used as differentials for testing Pst isolates (Hu et al., 2014; Wan et al., 2016). In China, a set of 19 wheat cultivars have been used as differentials to identify Pst races since the 1950s (Fang & Chen, 1955; Wan, Wu, Jin, et al., 2003). As the resistance genes of several Chinese differentials are unknown, race identification based on this set of differentials cannot directly reflect the avirulence/virulence of Pst to specific resistance genes, and the effectiveness of the resistance genes is hard to evaluate. For example, differential Zhong 4 is resistant to the currently predominant races CYR32 and CYR34 in China (Jia et al., 2022; Li et al., 2016; Liu & Chen, 2012; Wang, Zhan, et al., 2014; Wang, Cheng, et al., 2014) and is an important genetic stock for breeding stripe rust resistant cultivars. However, its genetic background and the resistance genes are not clear (Cao et al., 2013; Lin et al., 2007), and the avirulence genes in CYR32 and CYR34 have not been identified. Furthermore, many differentials in this set have two or more resistance genes, which may limit the genetic interpretation of their interactions with the avirulence genes of Pst. Therefore, it is necessary to establish a set of differentials with known resistance genes that can accurately reflect the virulence spectrum of the Pst populations; this should be helpful for deployment of resistance genes for the management of stripe rust.

The gene-for-gene hypothesis and the development of *Yr* near-isogenic lines (NILs) has proven to be an effective new strategy for Pst race identification (Flor, 1971; Wellings, 2007). It has been shown that the NILs with the Avocet S (AvS) background can effectively distinguish Pst races and so can directly monitor the effectiveness of stripe rust resistance genes (Wan, Wu, Jia, et al., 2003; Wellings, 2007). In 2010, US scientists began to use a set of differentials consisting of 18 *Yr* single-gene lines, and from this, established the PSTv nomenclature system for Pst races, which can accurately detect virulence factors to individual resistance genes (Wan et al., 2016; Wan & Chen, 2014). This set of differentials has been applied in many countries (Cat et al., 2023; Chen et al., 2021; Ghanbarnia et al., 2021), showing extensive ability to monitor the virulence composition of Pst populations worldwide. Although several studies have attempted to use this set of *Yr* single-gene differentials, it has not been systematically applied to monitor Pst races in China, limiting our understanding of virulence changes. The objectives of the present study were to identify Pst races from the 2023 collection throughout China using the set of 18 *Yr* single-gene differentials and to determine their virulence composition, frequency and geographical distribution. The results of this study can provide a theoretical basis for monitoring the virulence of Pst populations in China, guiding resistance breeding and rational deployment of stripe rust resistance genes.

2 | MATERIALS AND METHODS

2.1 | Collection of wheat stripe rust isolates

A field survey was conducted for the 2022/2023 wheat growing season, starting in south-western and north-western China in late October 2022, and in central and eastern China from March to June 2023. During the field survey, wheat leaves bearing Pst uredinia were collected (Figure 1, Table S1). The sampled infected leaves were air-dried at room temperature and kept in paper bags stored at 4°C for subsequent isolate purification and tests.

2.2 | Purification and multiplication of single-uredinium isolates

A 2-cm leaf segment bearing uredinia was cut from each sample and placed into a 2-mL centrifuge tube filled with Soltrol 170 isoparaffin (Chevron Phillips Chemical) so that urediniospores from the leaf

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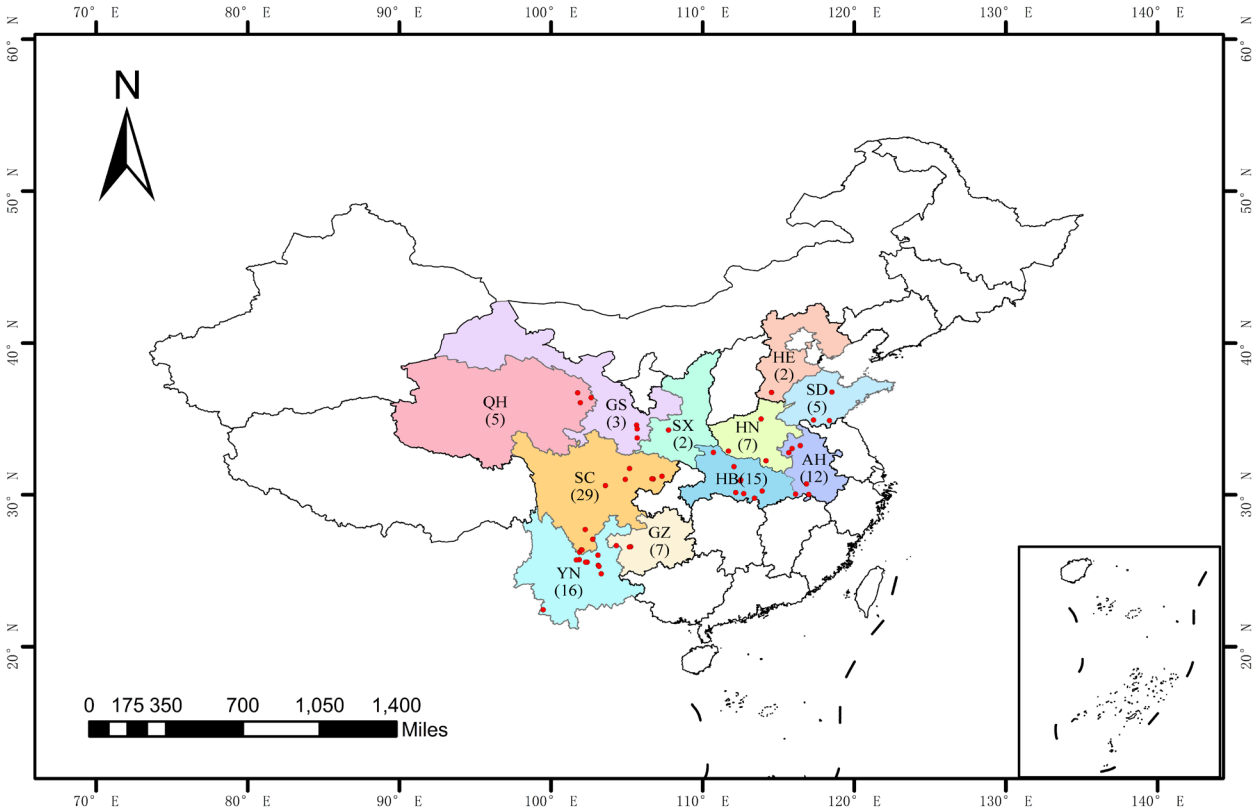

FIGURE 1 Map showing the distribution of *Puccinia striiformis* f. sp. *tritici* isolates collected in 2023 from 11 provinces in China. The number of isolates in each province is shown in parentheses and the red dots indicate the precise location of the sampling. The 11 provinces are AH, Anhui; GS, Gansu; GZ, Guizhou; HB, Hubei; HE, Hebei; HN, Henan; QH, Qinghai; SC, Sichuan; SD, Shandong; SX, Shaanxi; YN, Yunnan. [Colour figure can be viewed at wileyonlinelibrary.com]

segment were suspended in the Soltrol solution. The urediniospore suspension with final concentration of 50 mg/mL was used to spray over 7-day-old seedlings of the wheat cultivar Mingxian 169, which is susceptible to stripe rust, using a diaphragm vacuum pump (GM-0.5A, Jinteng Experimental Equipment). After leaving the inoculated plants in the room for 2 h to evaporate isoparaffin, they were transferred to an incubation room at 10°C, >90% humidity and no light. After incubation for 24 h, the inoculated plants were transferred to a greenhouse room maintained at 15–18°C during the 16-h light period and 11–15°C during the 8-h dark period. To prevent cross-contamination, each pot was covered with a glossy paper bag (40 × 30 cm). At 12–15 days after inoculation, the leaf segments with multiple uredinial pustules were removed to leave only one uredinial pustule on the leaf to obtain a single-uredinial isolate for each sample. Urediniospores from the single uredinial pustule were used to multiply spores by inoculating Mingxian 169 as described above. Urediniospores from these subsequent inoculations were collected by tapping the paper bag, dried on a filter paper in a Petri dish, and stored at 4°C until use.

2.3 | Virulence testing

The 18 *Yr* single-gene differentials established by Wellings et al. (2009) and Wan and Chen (2014) were used for identifying races of the Pst isolates. The set of differentials contains the genes *Yr1*, *Yr5*, *Yr6*, *Yr7*, *Yr8*, *Yr9*, *Yr10*, *Yr15*, *Yr17*, *Yr24*, *Yr27*, *Yr32*, *Yr43*, *Yr44*, *YrSP*, *Yr85* (*YrTr1*), *YrExp2* and *Yr76* (*YrTye*) (Feng et al., 2023; Wan & Chen, 2014; Xia et al., 2020; Xiang et al., 2016). AvS and Mingxian 169 were used as susceptible checks. Subsequently, the predominant races based on the *Yr* single-gene differentials were tested on the 19 Chinese differentials (Trigo-Eureka, Fulhard, Lutescens 128, Mentana, Virgilio, Abbondanza, Early Premium, Funo, Danish 1, Jubilejina II, Fengchan 3, Lovrin 13, Kangyin 655, Suwon 11, Zhong 4, Lovrin 10, Hybrid 46, *Triticum spelta* 'Album' and Guinong 22). Testing was performed as described below.

For each wheat line, seven seeds were planted in a pot (45 × 25 × 15 cm) filled with a soil mixture, and 8-day-old seedlings were inoculated with urediniospores as described above. Sixteen to 18 days after of inoculation, infection type (IT) scores of the first leaves were recorded using the 0–9 scale described by Line and Qayoum. (1992). The IT data were converted to avirulence/virulence: an isolate was defined as avirulent to a single *Yr* gene line with IT score of 0–6, or as virulent with IT score of 7–9 (Wan & Chen, 2014). The race for each isolate was identified according to its virulence/avirulence on the standard 18 *Yr* single-gene differentials using the US PSTv race series and the corresponding octal code of each race was assigned as proposed previously by Wan and Chen (2014) and modified by Wan et al. (2016). To distinguish from the PSTv races reported previously (Chen et al., 2002; Wan et al., 2016; Wan & Chen, 2014; Wan, Wang, et al., 2017; Wang et al., 2022), the races

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from China (CN) identified in this study using the 18 *Yr* single-gene differentials were named numerically using the same nomenclature of *P. striiformis* f. sp. *tritici* races, but with the preface PstCN.

2.4 | Virulence frequency, distribution, diversity and association

The Virulence Analysis Tool (VAT) (Kosman, 2014) was used to analyse the frequency, diversity and correlation among avirulence/virulence to *Yr* genes based on the isolates' virulence to tested wheat lines. The geographical distribution of virulence factors was analysed by calculating the percentage of each virulence factor in each province.

2.5 | Cluster analysis

According to the virulence or avirulence of each isolate on the 18 *Yr* single-gene differentials, the Nei genetic distance obtained from the VAT analysis was used to determine the genetic distance between isolates and between provincial populations of Pst. The 103 isolates and 11 provincial populations were clustered based on the avirulence/virulence data using the unweighted pair-group method of arithmetic means (UPGMA).

3 | RESULTS

3.1 | Race identification

A total of 157 wheat stripe rust samples were collected from 11 provinces in China in 2022/2023, from which 103 isolates were obtained. The details of the sample collection locations are given in Table S1 and Figure 1. A summary of the samples and isolates obtained from the 11 provinces is given in Table 1. The IT scores for all isolates are provided in Table S2 and Figure S1. The majority of isolates had distinct virulence phenotypes, either highly avirulent (with ITs of 0–2, 45.5%) or highly virulent (with ITs of 7–9, 51.4%). A total of 61 races were identified from the 103 isolates with a race:isolate ratio of 0.59, of which only 15 races were detected at least twice (Table 2). These most frequently found races were PstCN-043 (with a frequency of 8.7%), PstCN-008 (5.8%), PstCN-001 (4.8%), PstCN-036 (4.8%), PstCN-040 (3.9%), PstCN-045 (3.9%), PstCN-003 (2.9%), PstCN-005 (2.9%), PstCN-006 (2.9%), PstCN-014 (2.9%), PstCN-046 (2.9%) and PstCN-047 (2.9%). The remaining 46 races were represented by only one isolate. PstCN-008 (virulent to *Yr6*, *Yr8*, *Yr17*, *Yr27*, *Yr43*, *Yr44* and *YrExp2*) was the most widely distributed in Henan, Shandong, Shaanxi, Gansu, Sichuan and Anhui provinces and PstCN-043 (virulent to *Yr1*, *Yr6*, *Yr7*, *Yr8*, *Yr9*, *Yr17*, *Yr27*, *Yr43*, *Yr44* and *YrExp2*) was distributed in Yunnan, Guizhou, Sichuan and Anhui provinces. The numbers of virulence factors of the 61 races ranged from 3 to 16, with an average of 9.5 (Table 2). PstCN-022 from Shandong had the narrowest virulence spectrum (virulent only to *Yr1*, *Yr6* and *Yr7*). PstCN-061 from Hubei had the widest virulence spectrum (virulent to all *Yr* genes except *Yr5* and *Yr15*). Roughly, half of the isolates and races were virulent to at least nine out of 18 resistance genes.

3.2 | Virulence frequency, distribution and diversity

The Pst isolates had high frequencies (>40%) of virulence to *Yr1* (68.0%), *Yr6* (89.3%), *Yr7* (47.6%), *Yr8* (95.1%), *Yr9* (69.9%), *Yr17* (79.6%), *Yr27* (69.9%), *Yr43* (93.2%), *Yr44* (95.1%) and *YrExp2* (92.2%); moderate frequencies (10%–40%) to, *Yr24* (15.5%), *Yr32* (16.5%), *YrSP* (37.9%) and *Yr76* (37.9%); and low frequencies (<10%) to *Yr10* (8.7%) and *Yr85* (7.8%) (Figure S2). No isolates were virulent to *Yr5* or *Yr15*, indicating that these two genes were still effective in China.

TABLE 1 The number of samples collected, isolates recovered, and races of *Puccinia striiformis* f. sp. *tritici* identified from 11 provinces in China in 2023.

Province	No. of regions	No. of samples	No. of isolates	Recovery rate (%)	No. of races	Race:isolate ratio
HE	1	5	2	40.0	2	1
HN	3	12	7	58.3	7	1
SD	4	11	5	45.5	5	1
SX	1	5	2	40.0	2	1
SC	7	33	29	87.9	21	0.72
GS	2	10	3	30.0	3	1
QH	3	15	5	33.3	5	1
AH	4	18	12	66.7	11	0.92
GZ	2	8	7	87.5	6	0.92
HB	5	18	15	83.3	15	1
YN	4	22	16	72.7	11	0.69
Total (mean)	36	157	103	58.7	61	0.59

Abbreviations: AH, Anhui; GS, Gansu; GZ, Guizhou; HB, Hubei; HE, Hebei; HN, Henan; QH, Qinghai; SC, Sichuan; SD, Shandong; SX, Shaanxi; YN, Yunnan.

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TABLE 2 Races of *Puccinia striiformis* f. sp. *tritici* and their octal codes, virulence patterns, numbers, frequencies and distributions in China in 2023.

Race	PSTv[a]	Octal code	Virulence to *Yr* genes	No. of virulence factors	No. of isolates	Frequency (%) ($n = 103$)	Province (number of isolates)
PstCN-001	PSTv-136	020062	8, 43, 44, Exp2	4	5	4.8	SC (4), GZ (1)
PstCN-002	\	020262	8, 27, 43, 44, Exp2	5	1	1.0	QH (1)
PstCN-003	\	120062	6, 8, 43, 44, Exp2	5	3	2.9	HB (1), GS (1), AH (1)
PstCN-004	\	120063	6, 8, 43, 44, Exp2, 76	6	1	1.0	HB (1)
PstCN-005	\	120262	6, 8, 27, 43, 44, Exp2	6	3	2.9	AH (2), SC (1)
PstCN-006	\	121062	6, 8, 17, 43, 44, Exp2	6	3	2.9	SX (1), HN (1), SC (1)
PstCN-007	\	121066	6, 8, 17, 43, 44,85, Exp2	7	1	1.0	SC (1)
PstCN-008	\	121262	6, 8, 17, 27, 43, 44, Exp2	7	6	5.8	SD (1), SX (1), HN (1), GS (1), AH (1), SC (1)
PstCN-009	\	130262	6, 8, 9, 27, 43, 44, Exp2	7	1	1.0	HE (1)
PstCN-010	\	131262	6, 8, 9, 17, 27, 43, 44, Exp2	8	1	1.0	AH (1)
PstCN-011	\	131263	6, 8, 9, 17, 27, 43, 44, Exp2, 76	9	1	1.0	HB (1)
PstCN-012	\	131273	6, 8, 9, 17, 27, 43, 44, SP, Exp2, 76	10	1	1.0	YN (1)
PstCN-013	PSTv-261	150061	6, 7, 9, 43, 44, 76	6	1	1.0	HB (1)
PstCN-014	PSTv-119	161262	6, 7, 8, 17, 27, 43, 44, Exp2	8	3	2.9	HB (1), HN (1), AH (1)
PstCN-015	\	170062	6, 7, 8, 9, 43, 44, Exp2	7	1	1.0	AH (1)
PstCN-016	\	171072	6, 7, 8, 9, 17, 43, 44, SP, Exp2	9	1	1.0	GZ (1)
PstCN-017	\	400072	1, 43, 44, SP, Exp2	5	1	1.0	SD (1)
PstCN-018	\	411311	1, 9, 17, 27, 32, SP, 76	7	1	1.0	QH (1)
PstCN-019	\	420062	1, 8, 43, 44, Exp2	5	1	1.0	HE (1)
PstCN-020	\	431201	1, 8, 9, 17, 27, 76	6	1	1.0	QH (1)
PstCN-021	\	471063	1, 7, 8, 9, 17, 43, 44, Exp2, 76	9	1	1.0	HB (1)
PstCN-022	\	501000	1, 6, 17	3	1	1.0	SD (1)
PstCN-023	\	520262	1, 6, 8, 27, 43, 44, Exp2	7	1	1.0	HB (1)
PstCN-024	\	521262	1, 6, 8, 17, 27, 43, 44, Exp2	8	1	1.0	YN (1)
PstCN-025	\	530061	1, 6, 8, 9, 43, 44, 76	7	1	1.0	HB (1)
PstCN-026	\	530262	1, 6, 8, 9, 27, 43, 44, Exp2	8	1	1.0	SC (1)
PstCN-027	\	531073	1, 6, 8, 9, 17, 43, 44, SP, Exp2, 76	10	1	1.0	SC (1)
PstCN-028	\	531076	1, 6, 8, 9, 17, 43, 44, SP, 85, Exp2	10	1	1.0	YN (1)
PstCN-029	\	531212	1, 6, 8, 9, 17, 27, SP, Exp2	8	1	1.0	AH (1)
PstCN-030	\	531213	1, 6, 8, 9, 17, 27, SP, Exp2, 76	9	1	1.0	SC (1)
PstCN-031	\	531233	1, 6, 8, 9, 17, 27, 44, SP, Exp2, 76	10	1	1.0	HN (1)
PstCN-032	\	531261	1, 6, 8, 9, 17, 27, 43, 44, 76	9	1	1.0	HB (1)
PstCN-033	PSTv-7	531262	1, 6, 8, 9, 17, 27, 43, 44, Exp2	9	2	1.9	SC (2)
PstCN-034	\	531263	1, 6, 8, 9, 17, 27, 43, 44, Exp2, 76	10	1	1.0	YN (1)
PstCN-035	\	531271	1, 6, 8, 9, 17, 27, 43, 44, SP, 76	10	1	1.0	YN (1)
PstCN-036	\	531273	1, 6, 8, 9, 17, 27, 43, 44, SP, Exp2, 76	11	5	4.8	YN (3), QH (1), SC (1)
PstCN-037	\	535662	1, 6, 8, 9, 10, 17, 24, 27, 43, 44, Exp2	11	1	1.0	HB (1)

(Continues)

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TABLE 2 (Continued)

Race	PSTv[a]	Octal code	Virulence to *Yr* genes	No. of virulence factors	No. of isolates	Frequency (%) ($n = 103$)	Province (number of isolates)
PstCN-038	\	535773	1, 6, 8, 9, 10, 17, 24, 27, 32, 43, 44, SP, Exp2, 76	14	2	1.9	GZ (1), YN (1)
PstCN-039	\	555773	1, 6, 7, 9, 10, 17, 24, 27, 32, 43, 44, SP, Exp2, 76	14	1	1.0	SC (1)
PstCN-040	\	571062	1, 6, 7, 8, 9, 17, 43, 44, Exp2	9	4	3.9	YN (2), HB (1), SC (1)
PstCN-041	\	571072	1, 6, 7, 8, 9, 17, 43, 44, SP, Exp2	10	1	1.0	SC (1)
PstCN-042	\	571073	1, 6, 7, 8, 9, 17, 43, 44, SP, Exp2, 76	11	1	1.0	SC (1)
PstCN-043	PSTv-76	571262	1, 6, 7, 8, 9, 17, 27, 43, 44, Exp2	10	9	8.7	SC (3), YN (3), GZ (2), AH (1)
PstCN-044	PSTv-11	571263	1, 6, 7, 8, 9, 17, 27, 43, 44, Exp2, 76	11	1	1.0	SC (1)
PstCN-045	PSTv-144	571272	1, 6, 7, 8, 9, 17, 27, 43, 44, SP, Exp2	11	4	3.9	SC (3), AH (1)
PstCN-046	PSTv-17	571273	1, 6, 7, 8, 9, 17, 27, 43, 44, SP, Exp2, 76	12	3	2.9	SC (1), SD (1), GS (1)
PstCN-047	PSTv-220	571277	1, 6, 7, 8, 9, 17, 27, 43, 44, SP, 85, Exp2, 76	13	3	2.9	SD (1), YN (1), QH (1)
PstCN-048	\	571362	1, 6, 7, 8, 9, 17, 27, 32, 43, 44, Exp2	11	1	1.0	AH (1)
PstCN-049	\	571372	1, 6, 7, 8, 9, 17, 27, 32, 43, 44, SP, Exp2	12	1	1.0	AH (1)
PstCN-050	\	571373	1, 6, 7, 8, 9, 17, 27, 32, 43, 44, SP, Exp2, 76	13	1	1.0	SC (1)
PstCN-051	\	571573	1, 6, 7, 8, 9, 17, 24, 32, 43, 44, SP, Exp2, 76	13	1	1.0	SC (1)
PstCN-052	\	571663	1, 6, 7, 8, 9, 17, 24, 27, 43, 44, Exp2, 76	12	1	1.0	SC (1)
PstCN-053	\	571672	1, 6, 7, 8, 9, 17, 24, 27, 43, 44, SP, Exp2	12	1	1.0	HN (1)
PstCN-054	\	571726	1, 6, 7, 8, 9, 17, 24, 27, 32, 44, 85, Exp2	12	1	1.0	HB (1)
PstCN-055	\	571762	1, 6, 7, 8, 9, 17, 24, 27, 32, 43, 44, Exp2	12	1	1.0	HN (1)
PstCN-056	\	571773	1, 6, 7, 8, 9, 17, 24, 27, 32, 43, 44, SP, Exp2, 76	14	1	1.0	GZ (1)
PstCN-057	\	571776	1, 6, 7, 8, 9, 17, 24, 27, 32, 43, 44, SP, 85, Exp2	14	1	1.0	HB (1)
PstCN-058	\	575763	1, 6, 7, 8, 9, 10, 17, 24, 27, 32, 43, 44, Exp2, 76	14	2	1.9	HN (1), GZ (1)
PstCN-059	\	575771	1, 6, 7, 8, 9, 10, 17, 24, 27, 32, 43, 44, SP, 76	14	1	1.0	HB (1)
PstCN-060	\	575773	1, 6, 7, 8, 9, 10, 17, 24, 27, 32, 43, 44, SP, Exp2, 76	15	1	1.0	YN (1)
PstCN-061	PSTv-51	575777	1, 6, 7, 8, 9, 10, 17, 24, 27, 32, 43, 44, SP, 85, Exp2, 76	16	1	1.0	HB (1)

Abbreviations: AH, Anhui; GS, Gansu; GZ, Guizhou; HB, Hubei; HE, Hebei; HN, Henan; QH, Qinghai; SC, Sichuan; SD, Shandong; SX, Shaanxi; YN, Yunnan.

[a]'PSTv' is the race code denoted in the US Pst and '\' means the race is not found in the United States (https://striperust.wsu.edu).

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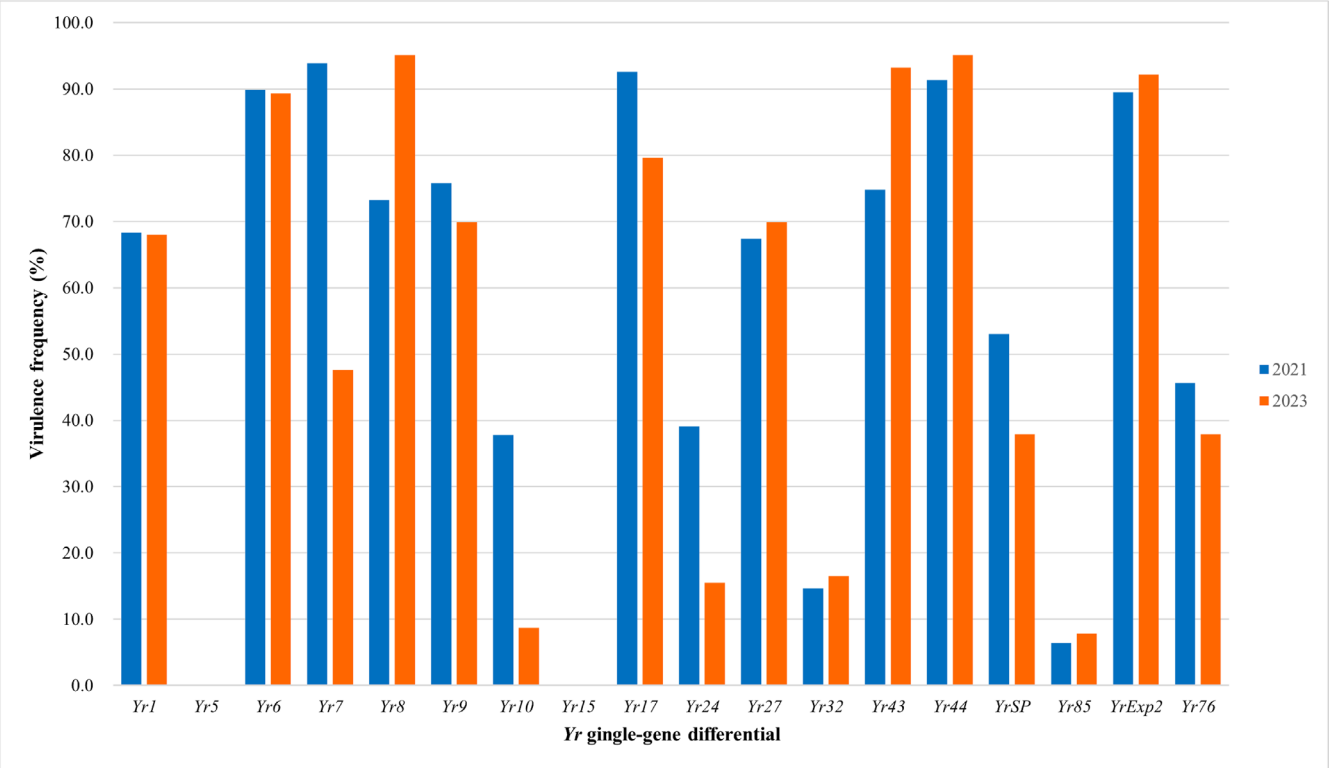

FIGURE 2 The virulence frequencies of *Puccinia striiformis* f. sp. *tritici* (Pst) to 18 *Yr* single-gene wheat differentials in China in 2021 and 2023. The data for virulence of Pst races in 2021 are taken from Zhou et al. (2023). [Colour figure can be viewed at wileyonlinelibrary.com]

Similar virulence frequencies were observed in Pst isolates from 2021, except that virulence factors to *Yr7*, *Yr10*, *Yr17*, *Yr24* and *YrSP* had higher frequencies in 2021 than in 2023 while those to *Yr8* and *Yr43* were higher in 2023 than 2021 (Figure 2; Table 3).

The distributions and frequencies of virulence factors varied among the provinces (Figure S3). The virulence spectra of the isolates in different provinces ranged from 7 to 16 of the *Yr* genes. The isolates from Shaanxi were only virulent to seven *Yr* genes, whereas the isolates from Sichuan, Yunnan and Hubei had the broadest virulence spectrum, virulent to all 18 *Yr* genes except *Yr5* and *Yr15*. Virulence factors to *Yr1*, *Yr6*, *Yr8*, *Yr27*, *Yr43*, *Yr44* and *YrExp2* were detected in all provinces. Virulence factors to *Yr10* and *Yr24* were detected in Henan, Sichuan, Guizhou, Hubei and Yunnan but not in Hebei, Shandong, Shaanxi, Gansu, Qinghai and Anhui (Table S3).

Slightly different virulence diversities of Pst were detected in different provinces (Table S4). The populations from Sichuan and Hubei had the highest diversity, and the lowest diversities were detected in the populations from Hebei and Shaanxi.

3.3 | Virulence associations

Highly positive correlations (correlation coefficient > 0.7) were observed for virulence factors to *Yr1* and *Yr9*, *Yr10* and *Yr24*, *Yr24* and *Yr32*, and *Yr43* and *Yr44*. Low and negative correlations (correlation coefficient < −0.1) were found for virulence factors to *Yr1* and *Yr43*, *Yr1* and *Yr44*, *Yr1* and *YrExp2*, *Yr8* and *Yr32*, *Yr8* and *YrSP*, *Yr8* and *Yr76*, *Yr9* and *YrExp2*, *Yr17* and *Yr43*, *Yr17* and *Yr44*, *Yr43* and *YrSP*, *Yr43* and *Yr76*, *Yr44* and *YrSP*, *Yr44* and *Yr76*, and *YrExp2* and *Yr76* (Table 4).

3.4 | Race clustering

Based on the virulence data analysed using the Nei's distances of 0.015 as the cut-off point, the 11 provincial Pst populations were clustered into seven groups (Figure 3). The Pst populations were similar in Guizhou, Henan and Hubei; Sichuan and Yunnan; and Anhui and Gansu. The populations in Shandong, Qinghai, Hebei and Shaanxi were distantly related to each other and to other provinces.

The 61 races were grouped into seven virulence groups (VGs) (Figure S4). VG1 consisted of 14 races that were virulent to *Yr1*, *Yr6*, *Yr7*, *Yr8*, *Yr9*, *Yr17*, *Yr43*, *Yr44* and *YrExp2*, sharing the first three digits of the octal codes 571 (Figure S4, Table 2). VG2 consisted of seven races, which were virulent to *Yr6*, *Yr8*, *Yr9*, Yr43, *Yr44* and *YrExp2*, sharing the fourth digit in the octal codes 0. VG3 consisted of 10 races that were virulent to *Yr6*, *Yr8*, *Yr9*, *Yr17* and *Yr27*, sharing the second to fourth digits of the octal code 312. VG4 had two races, PstCN-054 and PstCN-057. PstCN-054 was virulent to *Yr1*, *Yr6*, *Yr7*, *Yr8*, *Yr9*, *Yr17*, *Yr27*, *Yr32*, *Yr44*, *Yr85* and *YrExp2* and PstCN-057 had the same virulence spectrum plus virulence factors to *Yr43* and *YrSP*. VG5 consisted of seven races, all of which were virulent to *Yr10*, sharing the third digit of the octal codes 5. VG6 consisted of 18 races, all of which were virulent to *Yr43* and *Yr44* and avirulent to

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TABLE 3 The number of isolates and virulence frequencies to *Yr* genes of 13 races of *Puccinia striiformis* f. sp. *tritici* in China in 2021 and 2023.

		2021 (*N* = 608)[a]			2023 (*N* = 103)		
Octal code	Virulence to *Yr* gene	Name	No.	Frequency (%)	Name	No.	Frequency (%)
571273	1, 6, 7, 8, 9, 17, 27, 43, 44, SP, Exp2, 76	VP1	48	7.9	PstCN-046	3	2.9
571272	1, 6, 7, 8, 9, 17, 27, 43, 44, SP, Exp2	VP16	11	1.8	PstCN-045	4	3.9
571062	1, 6, 7, 8, 9, 17, 43, 44, Exp2	VP17	11	1.8	PstCN-040	4	3.9
571263	1, 6, 7, 8, 9, 17, 27, 43, 44, Exp2, 76	VP19	8	1.3	PstCN-044	1	1.0
571672	1, 6, 7, 8, 9, 17, 24, 27, 43, 44, SP, Exp2	VP23	6	1.0	PstCN-053	1	1.0
571262	1, 6, 7, 8, 9, 17, 27, 43, 44, Exp2	VP24	5	0.8	PstCN-043	9	8.7
575777	1, 6, 7, 8, 9, 10, 17, 24, 27, 32, 43, 44, SP, 85, Exp2, 76	VP27	5	0.8	PstCN-061	1	1.0
571073	1, 6, 7, 8, 9, 17, 43, 44, SP, Exp2, 76	VP39	3	0.5	PstCN-042	1	1.0
161262	6, 7, 8, 17, 27, 43, 44, Exp2	VP43	3	0.5	PstCN-014	3	2.9
170062	6, 7, 8, 9, 43, 44, Exp2	VP71	2	0.3	PstCN-015	1	1.0
531263	1, 6, 8, 9, 17, 27, 43, 44, Exp2, 76	VP78	1	0.2	PstCN-034	1	1.0
571072	1, 6, 7, 8, 9, 17, 43, 44, SP, Exp2	VP79	1	0.2	PstCN-041	1	1.0
575773	1, 6, 7, 8, 9, 10, 17, 24, 27, 32, 43, 44, SP, Exp2, 76	VP149	1	0.2	PstCN-060	1	1.0

[a]Data from 2021 were obtained from Zhou et al. (2023).

YrSP except for PstCN-017. VG7 had three races that were virulent to *Yr1* and *Yr17*, sharing the third digit of the octal codes 1.

4 | DISCUSSION

In this study, 103 Pst isolates collected from China in 2023 were characterized to 61 races using the differential set of 18 *Yr* single-gene wheat lines (Wan & Chen, 2014). The predominant races were PstCN-043 (octal code 571262) with frequency of 8.7% and PstCN-008 (octal code 121262) with frequency of 5.8%, which were detected in eight provinces.

With virulence to *Yr1*, *Yr6*, *Yr7*, *Yr8*, *Yr9*, *Yr17*, *Yr27*, *Yr43*, *Yr44* and *YrExp2* among the 18 *Yr* genes, PstCN-043 is equivalent to PSTv-76 in the United States. PSTv-76 was first discovered in Washington State in 2013, and subsequently detected in Montana in the same year and in Washington State in 2014 and 2015, in Idaho and North Carolina in 2016, and in Washington State again in 2018 and 2019. PSTv-76 has never become predominant in the United States, with a frequency of 0.7% among 1567 isolates tested from 2013 to 2017 (Wang et al., 2022), 1.3% in 2018 and 1.6% in 2019, and has not been detected in the United States since then (https://striperust.wsu.edu/races/data/). PSTv-76 was not detected among 491 isolates received from six countries over the period of 2013 to 2016 including 139 isolates from Yunnan, China, in 2016 (Chen et al., 2021). In a recent study, the virulence pattern (named VP24) of PstCN-043 or PSTv-76 was also identified in isolates from China in 2021, at a frequency of 80% in Liaoning (the total number of isolates is 17) and 20% in Yunnan (the total number of isolates is 30) but less than 1% across the country (Zhou et al., 2023). PSTv-76 was among the top four predominant races out of the 18 races identified from the stripe rust collections from Ethiopia in 2013 and 2014 (Wan, Muleta, et al., 2017). We noticed that race PstCN-043 belongs to a large virulence group sharing the virulence to *Yr1*, *Yr6*, *Yr7*, *Yr8*, *Yr9*, *Yr17*, *Yr43*, *Yr44* and *YrExp2* (octal code starting as 571). PstCN-046 from this study and PSTv-17 in the United States (Wan & Chen, 2014), also belong to the same virulence group, VG1, as PstCN-040. In fact, the PstCN-040 race group was also the most prevalent group in 2021 with the total frequency of 18.1%. Comparing the 2021 (Zhou et al., 2023) and 2023 race data, we found that 13 races were detected in both years. Surprisingly, these 13 shared races (again belongs to the PstCN-040 race group) were all virulent to *Yr6*, *Yr8*, *Yr44* and *YrExp2*, and have high virulence frequencies to *Yr1*, *Yr7*, *Yr9*, *Yr17* and *Yr43*, indicating that these resistance genes are no longer appropriate for wheat stripe rust resistance breeding alone. This PstCN-040 race group is currently prevalent in China, and continuous monitoring of the dynamics and evolution of this group should be a high priority.

In addition to the above-mentioned PstCN-043 (PSTv-76) and PstCN-046 (PSTv-17), 10 more races exist in both China and the United States, including PstCN-001, PstCN-013 and PstCN-014 (Table 2). Race PstCN-061 found in Hubei in 2023, with the widest virulence spectrum (virulent to all 18 *Yr* genes except *Yr5* and

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TABLE 4 The correlation coefficients of virulence (Vir) to 18 *Yr* genes determined by *Puccinia striiformis* f. sp. *tritici* isolates in China in 2023.

	Virulence to *Yr* gene														
	Yr6	*Yr7*	*Yr8*	*Yr9*	*Yr10*	*Yr17*	*Yr24*	*Yr27*	*Yr32*	*Yr43*	*Yr44*	*YrSP*	*Yr85*	*YrExp2*	*Yr76*
Yr1	0.167	0.404	−0.058	0.729	0.212	0.479	0.294	0.293	0.305	−0.185	−0.155	0.450	0.122	−0.122	0.364
Yr6		0.266	0.214	0.321	0.107	0.449	0.148	0.332	0.069	0.156	0.214	0.140	0.100	0.135	0.076
Yr7			0.034	0.498	0.118	0.386	0.289	0.140	0.310	0.180	0.215	0.178	0.159	0.131	0.058
Yr8				0.049	−0.090	0.110	−0.028	0.153	−0.143	0.298	0.369	−0.103	0.066	0.441	−0.103
Yr9					0.203	0.509	0.281	0.325	0.292	−0.093	−0.050	0.469	0.111	−0.111	0.469
Yr10						0.157	0.722	0.198	0.603	0.084	0.070	0.184	0.039	−0.039	0.326
Yr17							0.217	0.418	0.225	−0.137	−0.114	0.345	0.147	0.033	0.246
Yr24								0.216	0.748	0.009	0.097	0.218	0.176	0.024	0.273
Yr27									0.227	−0.088	−0.045	0.192	0.026	0.053	0.192
Yr32										−0.088	−0.021	0.300	0.164	−0.066	0.300
Yr43											0.836	−0.107	−0.066	0.354	−0.107
Yr44												−0.103	0.066	0.441	−0.103
YrSP													0.222	0.002	0.505
Yr85														0.084	0.073
YrExp2															−0.297

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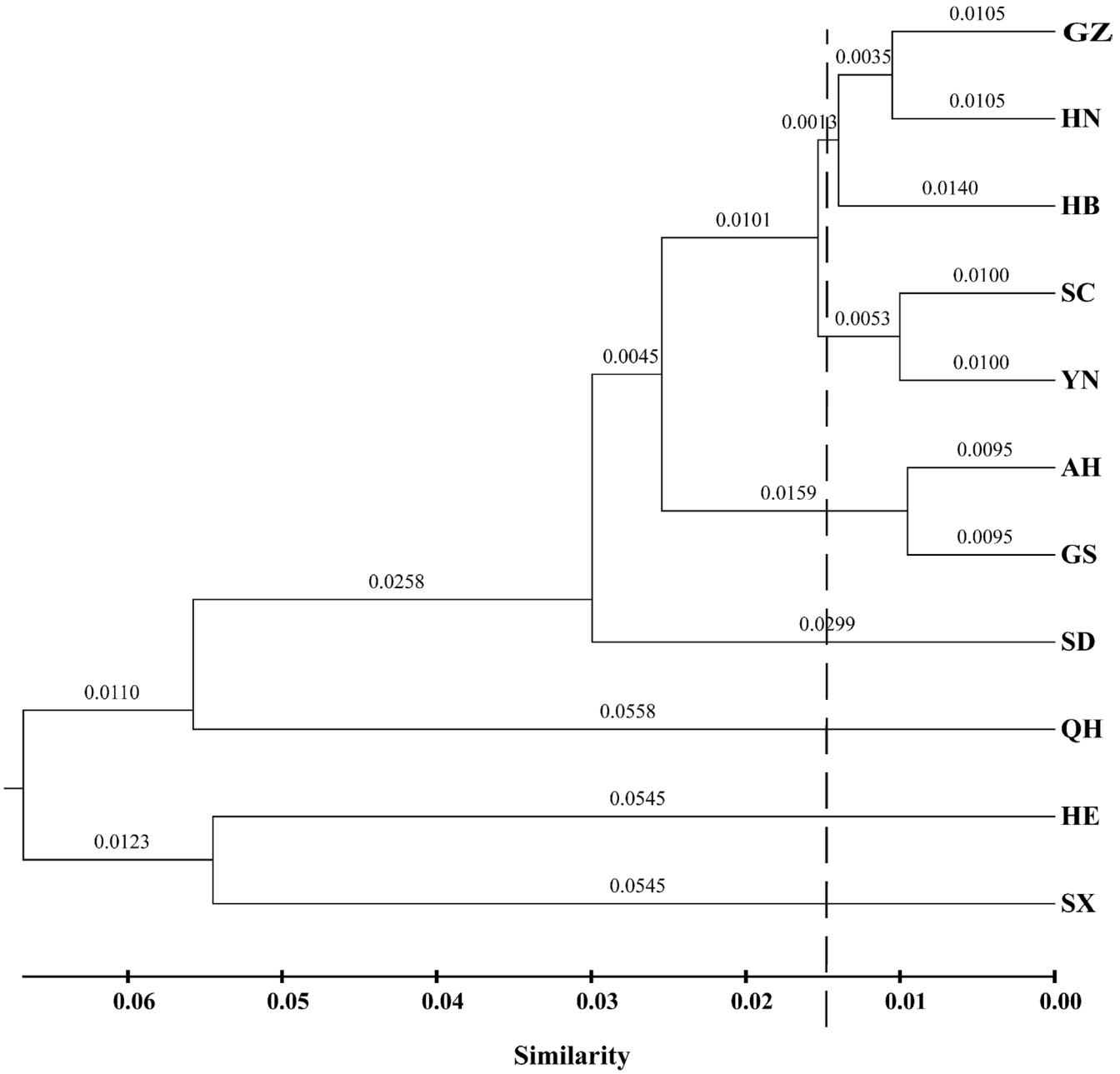

FIGURE 3 Dendrograms showing the relationships among *Puccinia striiformis* f. sp. *tritici* populations in 11 provinces of China in 2023 based on the virulence data on 18 *Yr* single-gene wheat differentials. The dashed line represents the Nei's distance cut-off point of 0.015. AH, Anhui; GS, Gansu; GZ, Guizhou; HB, Hubei; HE, Hebei; HN, Henan; QH, Qinghai; SC, Sichuan; SD, Shandong; SX, Shaanxi; YN, Yunnan.

Yr15), corresponds to PSTv-51 (octal code 575777), which was first identified in the United States in 2011 at a frequency of 0.6% (Wan et al., 2016). In a subsequent study, where historical US Pst isolates from 1968 to 2009 were tested, PSTv-51 was traced back to 2007 from only three isolates collected from 2007 to 2009 (Liu, Wan, et al., 2017). Since 2012, PSTv-51 has not been detected in the United States (Wan et al., 2016; Wang et al., 2022). In 2021, this race (termed as VP27) was also found in Henan and Shandong (Zhou et al., 2023). The detection of the same 10 races in both China and the United States suggest the worldwide spread of Pst isolates through long-distance air dispersal, human activities or possibly genetics (Boshoff et al., 2002), even though independent evolution could not be strictly excluded. Although the isolates of the same races need to be further studied using molecular markers or through genome sequencing to confirm their genetic relationships, these findings demonstrate the importance of using the same set of differentials to characterize Pst populations in different regions or countries.

China, due to its unique farming history, diverse climate conditions and strategic geographical location, stands as the world's largest independent epidemic region for wheat stripe rust. In 1988, Zeng introduced the concept of a regional epidemic system, dividing

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China's major stripe rust epidemic areas into 15 distinct regions to better understand and manage the disease's spread. Recent research has further elucidated the relationships between Pst populations across these regions. For example, Chen et al. (2013) classified the development of stripe rust into autumn sources of inoculum (including south of Gansu and Shaanxi provinces), spring sources of inoculum (including Sichuan, Yuannan and Guizhou provinces), and spring epidemic areas (including Henan, Hebei and Hubei provinces), highlighting the seasonal dynamics of the disease. Our study on Pst population clustering from various regions indicated that Pst populations from provinces like Sichuan and Yunnan formed a cluster, while those from Guizhou, Henan and Hebei comprised another. In particular, Hebei exhibited the highest virulence diversity index, supporting the hypothesis that it acts as a critical transit point for the pathogen's movement from western to eastern wheat-growing regions in China (Chen & Liu, 2023). This alignment with previous epidemic region classifications underscores the importance of continued surveillance and targeted management strategies to mitigate the impact of wheat stripe rust.

We attempted to compare the correspondence of the races identified from *Yr* single-gene lines and Chinese differentials. Nine isolates of the predominant PstCN-043 showed nine different virulence combinations on Chinese differentials (Table S5). Those isolates were virulent to Suwon11, therefore belonging to the Su11-virulence group. Combined with other races' virulence patterns on Chinese differentials (our unpublished data), it was found that the isolates of the same virulence group on Chinese differentials belonged to different races on the 18 *Yr* single-gene differentials, similar to the findings of Zhou et al. (2023). Because the resistance genes in the Chinese differentials are not accurately known, it is impossible to correctly postulate the composition of resistance genes through observing Pst–wheat interactions. However, the Chinese differentials have been used for more than 60 years. To study the virulence dynamics of the Pst population in China over time, new isolates should be continually tested with the Chinese differentials, as well as the *Yr* single-gene differentials, as in the previous study (Zhou et al., 2023). Alternatively, historical isolates that were tested with the Chinese differentials could be retested with the single-gene differentials, as performed by Liu, Wan, et al. (2017) on isolates in the United States.

Our study showed the different levels of effectiveness of 18 *Yr* genes in China. The frequencies of virulence to *Yr1*, *Yr6*, *Yr8*, *Yr9*, *Yr17*, *Yr27*, *Yr43*, *Yr44* and *YrExp2* were all higher than 60%, indicating that these resistance genes are largely ineffective in China and should be used cautiously in wheat resistance breeding. The results are consistent with previous reports. For example, circumvention of resistance of wheat cultivar Bima 1 (carrying *Yr1*) by the Pst populations caused the first reported destructive epidemics of stripe rust in China in the early 1960s (Li & Zeng, 2002; Xu et al., 2016). Trigo Eureka, carrying *Yr6*, was included in the Chinese set of differentials in 1967 because its resistance was overcome by the emergence of race CYR22 (Wang et al., 1986). *Yr8* was introduced into Chinese wheat varieties from *Aegilops comosa* in 1968 (Riley et al., 1968), but a race virulent to *Yr8* was found in 2013 and its frequency increased to 11.8% (Li et al., 2018). Wheat varieties carrying *Yr9* were introduced into China in the 1960s. By the end of the 1980s, 80% of wheat cultivars in China carried *Yr9*. However, in 1985, the emergence of race CYR29 overcame *Yr9* resistance (Chen et al., 2009; Wan et al., 2004). *Yr17*, derived from *Aegilops ventricosa*, has been used in northern European wheat varieties since the 1970s; although this gene exhibits strong additive effects with other minor genes at the adult plant stage in some studies (Ye et al., 2022), it has almost lost its effectiveness against most Pst races in the context of stripe rust in China (Coriton et al., 2020), as also shown in the present study. For this reason, *Yr17* could be still useful if introduced together with other resistance genes in breeding projects. *Yr44* was first identified from US spring wheat cultivar Zak (Cheng & Chen, 2010; Sui et al., 2009). As Zak (PI 607839) has a complex pedigree (Sui et al., 2009), the original source of *Yr44* has not been determined. Although Zak has not been widely used in wheat breeding programmes in China, the possibility that the origin of *Yr44* was in Chinese wheat landraces cannot be ruled out as Asian wheat landraces such as PI 167822 and Norin 10 were used to develop the parental lines of Zak. Nevertheless, high frequencies of *Yr44* virulence have been reported in many countries including China and the United States (Chen et al., 2021; Liu, Wan, et al., 2017; Sharma-Poudyal et al., 2013; Wan et al., 2016; Wan & Chen, 2014; Wan, Muleta, et al., 2017; Wang et al., 2022; Zhou et al., 2023). *YrExp2* was first identified in US spring wheat Express (PI 573003) (Lin & Chen, 2009), and the original donor of the gene is not clear. The high frequency of virulence to *YrExp2* found in our study is consistent with previous reports in China and other countries (Chen et al., 2021; Liu, Wan, et al., 2017; Sharma-Poudyal et al., 2013; Wan et al., 2016; Wan & Chen, 2014; Wan, Muleta, et al., 2017; Wang et al., 2022; Zhou et al., 2023).

Li et al. (2018) determined the virulence structure of the Pst population in Yunnan in 2016 using the same 18 *Yr* single-gene differentials from the United States and identified 64 races from 136 isolates. By comparing the present study with their results for Yunnan, we found that the virulence frequency against *Yr8* increased from 11.8% to 100%, for *Yr10* from 0 to 12.5%, for *Yr17* from 11.8% to 100%, for *Yr24* from 0.8% to 12.5%, for *Yr27* from 52.9% to 81.3%, and for *Yr32* from 0 to 12.5%. In contrast, the virulence frequencies against *Yr7* decreased from 91.9% to 81.6%, for *YrSP* from 80.1% to 43.8%, and for *Yr76* from 81.6% to 56.3%. These comparisons indicate that the Yunnan Pst population has changed greatly between 2016 and 2023. As an important oversummering zone, Yunnan is one of the places of origin of new Pst races in China (Xi et al., 2021); thus, continuous monitoring of virulence changes of the Pst populations in Yunnan is of great importance for wheat resistance breeding and rust management, not only for Yunnan, but also for other provinces. Li et al. (2023) monitored wheat diseases, including stripe rust, in Sichuan. They reported that from 2014 to 2022, the final disease severity on wheat carrying *Yr1*, *Yr6*, *Yr7*, *Yr8* and *Yr9* was ≥50% for nine consecutive years, indicating that these genes may be ineffective when used in wheat in Sichuan. Our study also showed that the frequency of virulence

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to these genes in Sichuan was greater than 50%. The virulence frequencies to the 18 *Yr* genes from our study in 2023 were similar to those found in 2021 by Zhou et al. (2023), except to *Yr10* (37.8% in 2021 vs. 8.7% in 2023) and *Yr24* (39.1% in 2021 vs. 15.5% in 2023). The high virulence to these genes in 2021 may have been due to the increase of race CYR34, which is virulent to *Yr10* and *Yr24* and became predominant by the epidemic in 2021 (Liu et al., 2022; Liu, Liu, et al., 2017; Zhou et al., 2023). Following the intensive control of stripe rust since 2021, the frequency of Pst isolates virulent to *Yr10* and *Yr24* decreased by 2023. Continually studying the Pst population dynamics should provide more detailed trends of the pathogen population evolution and its impact on the effectiveness of resistance genes. As the virulence frequencies of *Yr7*, *YrSP* and *Yr76* ranged from 37.9% to 47.6%, these genes should not be used as the main resistance genes in breeding. In contrast, *Yr10*, *Yr24*, *Yr32* and *Yr85* had low virulence frequencies of 7.8%–16.5%, and these genes could be used in combination with other effective resistance genes in breeding.

Although there are virulence factors with varying frequencies to the resistance genes discussed above, these genes can still play a crucial role in wheat breeding to enhance disease resistance and yield providing their use is combined with an understanding of virulence factors present in the pathogen population. First, it is essential to identify the key effective resistance genes such as *Lr34/Yr18/Pm38*, *Sr2* and *Fhb1* (Singh et al., 2005). Use of marker-assisted selection can significantly accelerate the breeding process by ensuring these genes are present in new varieties (Collard & Mackill, 2008). Gene pyramiding, which involves combining multiple resistance genes into a single variety, enhances the durability of disease resistance (Dendi et al., 2024). Additionally, understanding the mechanisms of virulence factors in pathogens, such as *Avr* genes and effector-triggered immunity, can help design more effective and targeted resistance strategies (Dodds & Rathjen, 2010). Employing gene-editing technologies such as CRISPR/Cas9 enables precise introduction or modification of resistance genes, allowing for greater control over the breeding process (Wang, Zhan, et al., 2014; Wang, Cheng, et al., 2014). It is also important to integrate these resistance genes with other essential agronomic traits, such as yield, drought tolerance and grain quality, to ensure that new varieties are well-rounded and meet broader agricultural needs (Reynolds & Borlaug, 2006). Continuous monitoring and adaptation of breeding strategies are necessary as pathogens evolve over time (McDonald & Linde, 2002). Finally, collaboration and data sharing with other researchers, institutions and international programmes can accelerate the breeding process and facilitate the development of more resilient, high-yielding and adaptable wheat varieties (Kosgey et al., 2006).

It is worth noting that no isolates virulent to *Yr5* and *Yr15* were detected in this study, consistent with previous studies of Pst collections from China and many other countries (Chen et al., 2021; Liu, Wan, et al., 2017; Sharma-Poudyal et al., 2013; Wan et al., 2016; Wan & Chen, 2014; Wan, Muleta, et al., 2017; Wang et al., 2022; Zhou et al., 2023). However, isolates virulent to *Yr5* have been occasionally reported in India (Nagarajan, 1986), Australia (Wellings & McIntosh, 1990), China (Zhang et al., 2020) and Turkey (Nagarajan, 1986; Tekin et al., 2021; Wellings & McIntosh, 1990; Zhang et al., 2020). Virulence to *Yr15* was reported only in Denmark and Afghanistan more than 30 years ago (van Silfhout, 1989) but has not been reported since. Thus, *Yr5* and *Yr15* still show excellent resistance to Pst worldwide (Cat et al., 2023; Chen et al., 2021; Ghanbarnia et al., 2021; Liu, Wan, et al., 2017; Sharma-Poudyal et al., 2013; Wamalwa et al., 2022; Wang et al., 2022) and these genes can be used together in breeding or in combination with other effective all-stage resistance genes, such as *Yr53*, *Yr64* and *Yr65* (Cheng et al., 2014; Qie et al., 2019; Xu et al., 2013). More importantly, numerous non-race-specific adult plant or high-temperature adult plant resistance genes, such as *Yr59*, *Yr62* and *Yr79* (Lu et al., 2014), should be used in combination with each other and with effective all-stage resistance genes for developing wheat cultivars with high level, durable resistance to stripe rust. It is important to continue monitoring the effectiveness of the all-stage resistance genes such as *Yr5* and *Yr15*. The predominant races and races with different virulence scores identified in the present study and in future race studies should be used to determine the effectiveness of these resistance genes, distinguish different types of resistance, evaluate breeding materials and discover new resistant germplasm and resistance genes.

In conclusion, 18 *Yr* single-gene differentials were used to characterize the virulence of 103 isolates from 11 provinces in China in 2023 and 61 races were identified. The dominant races were PstCN-040 and races sharing same virulence pattern. No isolates were detected that were virulent to *Yr5* or *Yr15*. The virulence frequencies to the other 16 *Yr* genes ranged from 7.8% to 95.1%. Continuous monitoring of Pst virulence will provide more details on the dynamics and evolution of Pst virulence, and further inform wheat breeding programmes for the use of effective resistance combinations to control this disease.

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DATA AVAILABILITY STATEMENT

The data that supports the findings of this study are available in the supplementary material of this article.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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